

	Click here to track your progress : Course Link								
			Youtube Link	Articles					
	1. Basics of programming								
	Problem Statement	Leetcode / GFG Link							
	1.1 Time and Space Complexity								
Pending	Learn about Time and Space Complexity	Theory [Watch the Video]	https://youtu.be/bLm0wgPU-6A	Time Complexity and Space Complexity in DSA Explained with Examples - Code and Debug - Blog					
Pending	What is TLE Error?	Theory [Watch the Video]	https://youtu.be/YwAnuOzOSY	-					
Pending	Time Complexity of Common Python Operations	Theory [Watch the Video]	https://youtu.be/PLV0f4RAMvI	-					
	1.2 Basic Maths Logic Buildup								
Pending	Extraction of Digits	Theory [Watch the Video]	https://youtu.be/yCMCdSZdlbY	Extraction of Digits in Python Explained Step-by-Step					
Pending	Count Digits	Number of Digits - Naukri Code 360	https://youtu.be/ww-mxKUItM	Python Program to Count Number of Digits Explained					
Pending	Reverse a number	Reverse Integer - LeetCode		Leetcode #7 : Reverse Integer Python Program Explained					
Pending	Check palindrome	Palindrome Number - LeetCode	https://youtu.be/ecZVpAZ37UI	Palindrome Number Program in Python Leetcode #9					
Pending	Armstrong Number	Armstrong Number - LeetCode	https://youtu.be/fzRwle8UwQQ	Python Program to Check Armstrong Number Explained					
Pending	Print all Divisors of a number	Number of factors Practice GeeksforGeeks	https://youtu.be/YoVlGYyWgyw	Python Program to Print Divisors/Factors of an Integer					
Pending	Check Prime	Prime Number Practice GeeksforGeeks	(Do on your own)	Check if a Number is Prime or Not GFG					
	1.3 Recursion Basics								
Pending	Introduction to Recursion	Theory [Watch the Video]	https://youtu.be/-VwidRe3Ob8	-					
Pending	Recursion using Parameters	Theory [Watch the Video]	https://youtu.be/fgurDxhawRw	-					
Pending	Functional Recursion	Theory [Watch the Video]	https://youtu.be/UgksZgiPrD4	-					
Pending	Print 1 to N without Loop	Print 1 To N Without Loop Practice GeeksforGeeks		Python Program to Print from 1 to N Without Loops					
Pending	Print N times with Recursion	Print GFG n times Practice GeeksforGeeks		Print N times with Recursion GFG					
Pending	Print N to 1 without Loop	Print N to 1 without loop Practice GeeksforGeeks		Print N to 1 without loop Normal Recursion and Backtracking					
Pending	Sum of first N natural numbers without Loop	Sum of first n terms Practice GeeksforGeeks		Sum of first n terms Using Recursion Explained					
Pending	Factorial of N numbers	Factorial Practice GeeksforGeeks	https://youtu.be/HKltpsIOOro	Python Program to Find Factorial of Number Using Recursion					
Pending	Reverse an Array without Loop	Reverse sub array Practice GeeksforGeeks	https://youtu.be/RynoW0e9I7U	Python Program to Reverse Array Using Recursion and While Loop					
Pending	Check if String is palindrome	Palindrome String Practice GeeksforGeeks	https://youtu.be/uWYF8nL5E1A	Palindrome String Recursive & Iterative Python Solutions					
Pending	Fibonacci Series	Fibonacci Number - LeetCode	https://youtu.be/33UUd-rkN9M	Fibonacci Number Recursive Python Solution					
	1.4 Hashing								
Pending	[Pre Requirement] How to store frequency	Theory [Watch the Video]	https://youtu.be/Ocd_uHpsQyg						
Pending	Learn basics of hashing	Theory [Watch the Video]	https://youtu.be/0IE5T6qEHg						
Pending	Find the Frequency	Find the Frequency Practice GeeksforGeeks	(Do on your own)	-					
Pending	Count Frequency in a range	Frequencies of Limited Range Array Elements Practice GeeksforGeeks	(Do on your own)	-					
	2. Different types of Sorting								
Pending	Selection Sort	Selection Sort Practice GeeksforGeeks	https://youtu.be/x8pZMsEivJk	Python Program for Selection Sort Algorithm					
Pending	Bubble Sort	Bubble Sort Practice GeeksforGeeks	https://youtu.be/M6Vg8SmGMA8	Bubble Sort Algorithm in Python Explained in Depth					
Pending	Insertion Sort	Insertion Sort Practice GeeksforGeeks	https://youtu.be/-Cl6Lhv1FOw	Insertion Sort Algorithm Explained in Python					
Pending	Merge Sort	Merge Sort Practice GeeksforGeeks	https://youtu.be/a2gP9uYrpQo	Merge Sort Algorithm in Python Explained					
Pending	Quick Sort	Quick Sort Practice GeeksforGeeks	https://youtu.be/77tOJ3kVeS0	Quick Sort Algorithm in Python Explained					
Pending	Recursive Bubble Sort	Bubble Sort Practice GeeksforGeeks	(Do on your own)	Recursive Bubble Sort in Python					
Pending	Recursive Insertion Sort	Insertion Sort Practice GeeksforGeeks	(Do on your own)	Recursive Insertion Sort in Python					
	3. Problems on Arrays								
	3.1 EASY								
Pending	Largest Element in an Array	Largest Element in Array Practice GeeksforGeeks	https://youtu.be/H52p90uVUGA	Find the Largest Element in an Array (Brute Force & Optimal Solutions) Step-by-Step with Images					
Pending	Second Largest Element in an Array without sorting	Second Largest Number - Naukri Code 360	https://youtu.be/Z6TwNJ39OoQ	Find the second largest and second smallest element in an Array Explained					
Pending	Check if the array is sorted	Check if array is sorted Practice GeeksforGeeks	https://youtu.be/_T4tR2Y6yd4	Check if array is sorted Explained using Python Code					
Pending	Remove duplicates from Sorted array	Remove Duplicates from Sorted Array - LeetCode	https://youtu.be/a2gP9uYrpQo	Remove duplicates from Sorted array - Leetcode 26					
Pending	Right Rotate an array by one place	Theory [Watch the Video]	https://youtu.be/_EHlC6xhhE0	-					
Pending	Right rotate an array by K places	Rotate Array - LeetCode	https://youtu.be/HRzQK5TkriZM	Right Rotate an Array by K places Explained with Examples					
Pending	Move Zeros to end	Move Zeroes - LeetCode	https://youtu.be/GtssSuFh_Ek	Move Zeroes Leetcode 283 Explained					
Pending	Linear Search	Array Search Practice GeeksforGeeks	https://youtu.be/mjijiUK0Z1E	Linear Search Explained in Python					
Pending	Merge 2 sorted Arrays	Union of Two Sorted Arrays Practice GeeksforGeeks	https://youtu.be/d9Rpk0j9ofs	Union of 2 Sorted with Duplicates Explained with Images					
Pending	Find missing number in an array	Missing Number - LeetCode	https://youtu.be/ZoId2EqX45U	Missing Number Leetcode 268 Explained					
Pending	Maximum Consecutive Ones	Max Consecutive Ones - LeetCode	https://youtu.be/1QLnjtdEI9Y	Maximum Consecutive Ones Leetcode 485 Explained with Images					
	3.2 MEDIUM								
Pending	2Sum Problem	Two Sum - LeetCode	https://youtu.be/EX2tR_cgCz4	https://codeanddebug.in/blog/two-sum-leetcode-1/					
Pending	Kadane's Algorithm, maximum subarray sum	Maximum Subarray - LeetCode	https://youtu.be/o3luQXt08W8	https://codeanddebug.in/blog/maximum-subarray-kadanes-algorithm/					
Pending	Stock Buy and Sell	Best Time to Buy and Sell Stock - LeetCode	https://youtu.be/GrDqGxWsmc	Best Time to Buy and Sell Stock Leetcode 121 Explained with Images					

Pending	Rearrange the array in alternating positive and negative items	Rearrange Array Elements by Sign - LeetCode	https://youtu.be/xFm1Tgc-HTI	Rearrange Array Elements by Sign Leetcode 2149
Pending	Longest Consecutive Sequence in an Array	Longest Consecutive Sequence - LeetCode	https://youtu.be/OO2FsBeFcEs	Longest Consecutive Sequence Leetcode 128
Pending	Learn 2D Matrix [Check video]	Learn 2D Matrix [Watch the Video]	https://youtu.be/sVzamPflrw	
Pending	Set Matrix Zeros	Set Matrix Zeroes - LeetCode	https://youtu.be/_RMrPINBftA	Set Matrix Zeroes Leetcode 73 Explained
Pending	Rotate Matrix by 90 degrees	Rotate Image - LeetCode	https://youtu.be/w9ofZwey57M	Rotate Image Leetcode 48 Explained with Images
Pending	Print the matrix in spiral manner	Spiral Matrix - LeetCode	https://youtu.be/FHnZAvZN-Co	Spiral Matrix Leetcode 54 Explained in Python
		3.3 HARD		
Pending	3-Sum Problem	3Sum - LeetCode	https://youtu.be/Euky4JTEK0I	3Sum Leetcode 15 Explained in Python
Pending	4-Sum Problem	4Sum - LeetCode	https://youtu.be/8DwLoo5AOR8	4Sum Leetcode 18 Explained in Python
		4. Binary Search		
		4.1 Binary Search on simple 1D Arrays		
Pending	Binary Search to find X in sorted array	Binary Search - LeetCode	https://youtu.be/fM4Vat0dERo	Binary Search Leetcode 704 Explained in Python
Pending	Implement Lower Bound	Floor in a Sorted Array Practice GeeksforGeeks	https://youtu.be/FTtbMO4o9OI	Implement Lower Bound Floor in a Sorted Array (GFG)
Pending	Implement Upper Bound	Implement Upper Bound - Naukri Code 360	https://youtu.be/FTtbMO4o9OI	Implement Upper Bound
Pending	Search Insert Position	Search Insert Position - LeetCode	https://youtu.be/MhsEFYylmPs	Search Insert Position Leetcode 35 Explained in Python
Pending	Floor/Ceil in Sorted Array	Ceil The Floor - Naukri Code 360	https://youtu.be/WlombWDEPJg	Ceil The Floor Binary Search Implementation
Pending	Find the first or last occurrence of a given number in a sorted array	Find First and Last Position of Element in Sorted Array - LeetCode	https://youtu.be/ViotQoXhip4	Find First and Last Position of Element in Sorted Array Leetcode 34 Optimal Binary Search
Pending	Count occurrences of a number in a sorted array with duplicates	Number of occurrence Practice GeeksforGeeks	https://youtu.be/Hs49-Lws_XM	Count occurrences of a number in a sorted array with duplicates Optimal Binary Search
Pending	Search in Rotated Sorted Array I	Search in Rotated Sorted Array - LeetCode	https://youtu.be/46kRvrlSxM	Search in Rotated Sorted Array Leetcode 33 Optimal Binary Search
Pending	Search in Rotated Sorted Array II	Search in Rotated Sorted Array II - LeetCode	https://youtu.be/oVcBXszcpYs	Search in Rotated Sorted Array II Leetcode 81 Optimal Binary Search
Pending	Find minimum in Rotated Sorted Array	Find Minimum in Rotated Sorted Array - LeetCode	https://youtu.be/_B2JakopXLC	Find Minimum in Rotated Sorted Array Leetcode 153 Optimal Binary Search
		5. Strings (please do at last) (It might contain other patterns thats why)		
		5.1 EASY		
Pending	Remove outermost Paranthesis	Remove Outermost Parentheses - LeetCode	https://youtu.be/yjs09jkYPbs	Remove Outermost Parentheses Leetcode 1021 Stack Depth Counting Solution
Pending	Reverse words in a given string	Reverse Words in a String - LeetCode	https://youtu.be/EBd-u0ehi98	Reverse Words in a String Leetcode 151 Split and Reverse Approach
Pending	Largest odd number in a string	Largest Odd Number in String - LeetCode	https://youtu.be/kOiapvtj0I	Largest Odd Number in String Leetcode 1903 Step-by-Step Solution Explained
Pending	Longest Common Prefix	Longest Common Prefix - LeetCode	https://youtu.be/2iY7YYLNLAA	Longest Common Prefix Leetcode 14 Optimal Solution Explained
Pending	Isomorphic String	Isomorphic Strings - LeetCode	https://youtu.be/cX9vCdJlNyBo	Isomorphic Strings Leetcode 205 Optimal Hash Map Solution
Pending	Check whether one string is a rotation of another	Rotate String - LeetCode	https://youtu.be/PrSGLUfmLh0	Rotate String Leetcode 796 Optimal Substring Search Solution
Pending	Check if two strings are anagram of each other	Valid Anagram - LeetCode	https://youtu.be/EnrE4tmYZo	Valid Anagram Leetcode 242 Optimal Hash Map Solution
		5.2 MEDIUM		
Pending	Sort Characters by frequency	Sort Characters By Frequency - LeetCode	https://youtu.be/pyOb7lwnA7M	Sort Characters By Frequency Leetcode 451 Hash Map + Sorting Solution
Pending	Maximum Nesting Depth of Paranthesis	Maximum Nesting Depth of the Parentheses - LeetCode	https://youtu.be/5XgWK4Dj9UY	Maximum Nesting Depth of the Parentheses Leetcode 1614 Depth Counting Solution
Pending	Roman Number to Integer and vice versa	Roman to Integer - LeetCode	https://youtu.be/1a7oYSZtCaY	Roman to Integer Leetcode 13 Mapping and Subtraction Rule Solution
		6. Linked List (Single LL, Double LL)		
		6.1 Learn Singly Linked List		
Pending	Introduction to LinkedList	Introduction to Linked List Practice GeeksforGeeks	https://youtu.be/-1WwCkUEMO	
Pending	Inserting a node in LinkedList	Linked List Insertion Practice GeeksforGeeks		
Pending	Deleting a node in LinkedList	Delete Node in a Linked List - LeetCode		Understanding Singly Linked List in Python Complete Explanation and Implementation
Pending	Find the length of the linkedlist	Count nodes of linked list Practice GeeksforGeeks	https://youtu.be/h7w9U7w8JfE	
Pending	Search an element in the LL	Search in Linked List Practice GeeksforGeeks		
Pending	Design Linked List	Design Linked List - LeetCode		Design Linked List Leetcode 707 Explained in Python
		6.2 Learn Doubly Linked List		
Pending	Introduction to DLL	Introduction to Doubly Linked List Practice GeeksforGeeks	https://youtu.be/wiMmDsFHna8	
Pending	Insert a node in DLL	Doubly linked list Insertion at given position Practice GeeksforGeeks		
Pending	Delete a node in DLL	Delete node in Doubly Linked List Practice GeeksforGeeks	https://youtu.be/41q73CpGnc	Understanding Doubly Linked Lists: Complete Guide with Real-Life Examples
Pending	Reverse a DLL	Reverse a Doubly Linked List Practice GeeksforGeeks	https://youtu.be/G1QIY6YbDA	Reverse a Doubly Linked List GFG Practice Iterative Solution
		6.3 Medium problems of SLL		
Pending	Middle of a LinkedList [TortoiseHare Method]	Middle of the Linked List - LeetCode	https://youtu.be/7YEEZjO03JwA	Middle of the Linked List Leetcode 876 Tortoise Hare Approach
Pending	Reverse a LinkedList [Iterative]	Reverse Linked List - LeetCode	https://youtu.be/8DJwQV3fjUl	Reverse Linked List Leetcode 206 Iterative vs Recursive Solution
Pending	Detect a loop in LL	Linked List Cycle - LeetCode	https://youtu.be/2MIUoKhJjA0	Linked List Cycle Leetcode 141 Floyd's Cycle Detection
Pending	Find the starting point in LL	Linked List Cycle II - LeetCode	https://youtu.be/8qlar5RrRic	Linked List Cycle II Leetcode 142 Complete Guide with 2 Different Solutions
Pending	Length of Loop in LL	Find length of Loop Practice GeeksforGeeks	https://youtu.be/cCaLT3un9VE	Find length of Loop Complete Guide with 2 Different Solutions
Pending	Segregate odd and even nodes in LL	Odd Even Linked List - LeetCode	https://youtu.be/DFNPTtevXso	Odd Even Linked List Leetcode 328 Brute to Optimal Solution
Pending	Remove Nth node from the back of the LL	Remove Nth Node From End of List - LeetCode	https://youtu.be/L7_0FCrajUA	Remove Nth Node From End of List Leetcode 19
		6.4 Medium problems of DLL		

[illegible]

Pending	Asteroid Collision	Asteroid Collision - LeetCode	https://youtu.be/VLw3CaRz_aQ	Asteroid Collision Leetcode 735 Optimal Stack Solution					
	10. Sliding Window and Two Pointers								
	10.1 Medium level Problems								
Pending	Longest Substring Without Repeating Characters	Longest Substring Without Repeating Characters - LeetCode	https://youtu.be/_zpYpeIRePE	https://codeanddebug.in/blog/longest-substring-without-repeating-characters/					
Pending	Max Consecutive Ones III	Max Consecutive Ones III - LeetCode	https://youtu.be/VgmNnpc5ycQ	Max Consecutive Ones III Leetcode 1004 Fixed Window Solution					
Pending	Fruit Into Baskets	Fruit Into Baskets - LeetCode	https://youtu.be/bxMYNPMrwo	Fruit Into Baskets Leetcode 904 Optimal Solution using Sliding Window					
Pending	Maximum point you can obtain from cards	Maximum Points You Can Obtain from Cards - LeetCode	https://youtu.be/SuwXaaXeOaO	Maximum Points You Can Obtain from Cards Leetcode 1423					
	11. Greedy Algorithms								
	11.1 Easy Level Problems								
Pending	Assign Cookies	Assign Cookies - LeetCode	https://youtu.be/zboVRO7xv-0	Assign Cookies Leetcode 455 Optimal Greedy Solution					
Pending	Fractional Knapsack Problem	Fractional Knapsack Practice GeeksforGeeks	https://youtu.be/K_qtZbLSym0	https://codeanddebug.in/blog/fractional-knapsack/					
Pending	Greedy algorithm to find minimum number of coins	Minimum number of Coins Practice GeeksforGeeks	https://youtu.be/hMcQXv3pT2o	Minimum number of Coins Greedy Algorithm Solution					
Pending	Lemonade Change	Lemonade Change - LeetCode	https://youtu.be/OtxKdaDnEOE	Lemonade Change Leetcode 860 Greedy Solution					
	11.2 Medium/Hard Level Problems								
Pending	N meetings in one room	N meetings in one room Practice GeeksforGeeks	https://youtu.be/OnQIEihg1sA	N meetings in one room Greedy Solution					
Pending	Jump Game	Jump Game - LeetCode	https://youtu.be/kw8mliTKLK4	Jump Game Leetcode 55 Greedy Solution					
Pending	Jump Game 2	Jump Game II - LeetCode	https://youtu.be/H4T4v4-yi9m8	Jump Game II Leetcode 45 Recursive and Greedy Solution					
Pending	Minimum number of platforms required for a railway	Minimum Platforms Practice GeeksforGeeks	https://youtu.be/z5yb0JGi4OE	Minimum Platforms Optimal Greedy Solution					
	12. Binary Trees								
	12.1 Learn Traversal								
Pending	Introduction to Trees	Introduction to Trees Practice GeeksforGeeks	https://youtu.be/zquPPfnD87U						
Pending	Create Binary Tree	Binary Tree Representation Practice GeeksforGeeks	https://youtu.be/ap2ctlcOgdM						
Pending	Binary Tree Traversals in Binary Tree	Tree Traversals - Naukri Code 360							
Pending	Preorder Traversal of Binary Tree	Binary Tree Preorder Traversal - LeetCode	https://youtu.be/ev03rFgPO4E	A Beginner's Guide to Preorder, Inorder & Postorder Traversals					
Pending	Inorder Traversal of Binary Tree	Binary Tree Inorder Traversal - LeetCode							
Pending	Post-order Traversal of Binary Tree	Binary Tree Postorder Traversal - LeetCode							
Pending	Level order Traversal / Level order traversal in spiral form	Binary Tree Level Order Traversal - LeetCode	https://youtu.be/tRhWzxU-oH0	Breadth-First Search in Binary Trees: Level-Order Traversal with Python					
	12.2 Medium Level Problems								
Pending	Height of a Binary Tree	Maximum Depth of Binary Tree - LeetCode	https://youtu.be/qYMXdq2abD8	Maximum Depth of Binary Tree Explained with Examples					
Pending	Check if the Binary tree is height-balanced or not	Balanced Binary Tree - LeetCode	https://youtu.be/vjZ8A7mBOY	Balanced Binary Tree Leetcode 110 Optimal and Slightly Optimal Solutions					
Pending	Diameter of Binary Tree	Diameter of Binary Tree - LeetCode	https://youtu.be/KsAdE9oiKIA	Diameter of Binary Tree					
Pending	Maximum path sum	Binary Tree Maximum Path Sum - LeetCode	https://youtu.be/kpEWBJT3gbl	Binary Tree Maximum Path Sum Leetcode 124 Detailed Recursive Solution					
Pending	Top View of Binary Tree	Top View of Binary Tree Practice GeeksforGeeks	https://youtu.be/g8cmTH2pCj8	Top View of Binary Tree BFS with Horizontal Distance Mapping					
Pending	Bottom View of Binary Tree	Bottom View of Binary Tree Practice GeeksforGeeks	https://youtu.be/KRMSZ8-vQcE	Bottom View of Binary Tree BFS with Horizontal Distance Mapping					
Pending	Right/Left View of Binary Tree	Binary Tree Right Side View - LeetCode	https://youtu.be/9ldm6RRruhM	Binary Tree Right Side View Leetcode 199 BFS and DFS Approaches					
	13. Binary Search Trees								
	13.1 Introduction								
Pending	Introduction to Binary Search Tree	Binary Search Trees Practice GeeksforGeeks	https://youtu.be/3Ig6lkNIEik						
Pending	Search in a Binary Search Tree	Search in a Binary Search Tree - LeetCode	https://youtu.be/hF3EcB3grg8						
Pending	Find Min/Max in BST	Minimum element in BST Practice GeeksforGeeks	https://youtu.be/GIHS4wevMFU						
	13.2 Practice Problems								
Pending	Ceil in a Binary Search Tree	Ceil in BST Practice GeeksforGeeks	https://youtu.be/GIHwR_vsm4						
Pending	Floor in a Binary Search Tree	Floor in BST Practice GeeksforGeeks	https://youtu.be/wEaEmxyYDNs						
Pending	Insert a given Node in Binary Search Tree	Insert into a Binary Search Tree - LeetCode	https://youtu.be/CldBMbj_33c						
Pending	Delete a Node in Binary Search Tree	Delete Node in a BST - LeetCode	https://youtu.be/GKvN-MISnKE						
Pending	Morris Preorder/Inorder Traversal of a Binary Tree	Binary Tree Inorder Traversal - LeetCode	https://youtu.be/jwVhYBDVUQ0						
Pending	Find K-th smallest/largest element in BST	Kth Smallest Element in a BST - LeetCode	https://youtu.be/g4SXMP-BUkk						
Pending	Check if a tree is a BST or BT	Validate Binary Search Tree - LeetCode	https://youtu.be/ah604p9o5rA						
Pending	LCA in Binary Search Tree	Lowest Common Ancestor of a Binary Search Tree - LeetCode	https://youtu.be/huVYorR9rxE						
Pending	Inorder Successor/Predecessor in BST	Inorder Successor in BST - LeetCode	https://youtu.be/-PHBkE0u9GA						
	14. Heaps (do after binary trees)								
	14.1 Introduction to Heaps								
Pending	Introduction to Priority Queues using Binary Heaps	Implementation of Priority Queue using Binary Heap Practice GeeksforGeeks	https://youtu.be/iUNcm2PgZPU						
Pending	Heapify Algorithm	-	https://youtu.be/ULv4UGQi-1U						
Pending	Build Heap from a Given Array	-	https://youtu.be/dTVie190jU						
Pending	Implement Min Heap	-	https://youtu.be/EqOTFRdsLOC						

Pending	Implement Max Heap		https://youtu.be/xdy3XyIeyE		
Pending	Check if an Array Represents a Min Heap	-	https://youtu.be/AxcEJLQBELU		
Pending	Convert Min Heap to Max Heap	-	https://youtu.be/VHdKsoRft8U		
		14.2 Medium level problems			
Pending	Kth largest element in an array [use priority queue]	Kth Largest Element in an Array - LeetCode			
Pending	Kth smallest element in an array [use priority queue]	Kth smallest element Practice GeeksforGeeks			
Pending	Merge M sorted Lists	Merge k Sorted Lists - LeetCode			
Pending	Replace each array element by its corresponding rank	Replace elements by its rank in the array Practice GeeksforGeeks			
Pending	Task Scheduler	Task Scheduler - LeetCode			
Pending	Hands of Straights	Hand of Straights - LeetCode			
		15. Graphs			
		15.1 Introduction to Graphs			
Pending	Graph and Types	Graph and Vertices Practice GeeksforGeeks	https://youtu.be/0IVnSenyoRU	[Watch Video]	
Pending	Graph Representation	Print adjacency list Practice GeeksforGeeks	https://youtu.be/Y54FmJBHHDl	[Watch Video]	
Pending	Connected Components	Number of Connected Components in an Undirected Graph - LeetCode	https://youtu.be/YSR0dEqf3ow	[Watch Video]	
Pending	BFS	BFS of graph Practice GeeksforGeeks	https://youtu.be/lxOw5e7ZuVg	BFS Traversal in Graph Explained using Code	
Pending	DFS	DFS of Graph Practice GeeksforGeeks	https://youtu.be/eqch9gKiWEQ	DFS Traversal in Graph Explained using Code	
		15.2 Problems based on BFS & DFS			
Pending	Rotten Oranges	Rotting Oranges - LeetCode	https://youtu.be/s8BspS2NU7Oc	Rotting Oranges Leetcode 994 Explained	
Pending	Flood fill	Flood Fill - LeetCode	https://youtu.be/nisR9Nyt4ww4	Flood Fill Leetcode 733 DFS and BFS Approach	
Pending	Cycle Detection in undirected Graph (bfs)	Detect cycle in an undirected graph Practice GeeksforGeeks	https://youtu.be/k7Hd32SiRGo	Detect cycle in an undirected graph using BFS	
Pending	Cycle Detection in undirected Graph (dfs)	Detect cycle in an undirected graph Practice GeeksforGeeks	https://youtu.be/XUKV0_AFh44	Detect cycle in an undirected graph using DFS	
Pending	0/1 Matrix (Bfs Problem)	01 Matrix - LeetCode	https://youtu.be/p-om5wqOjal	01 Matrix Leetcode 542 Explained using BFS	
Pending	Surrounded Regions (dfs)	Surrounded Regions - LeetCode	https://youtu.be/n05WmMKTcCo	Surrounded Regions Leetcode 130 Explained using DFS	
Pending	Number of Enclaves [flood fill implementation - multisource]	Number of Enclaves - LeetCode	https://youtu.be/5cDdI2Z2ToI	Number of Enclaves Leetcode 1020 Explained using BFS	
Pending	Word ladder - 1	Word Ladder - LeetCode	https://youtu.be/hxZQY5U__NM	Word Ladder Leetcode 127 Explained using BFS	
Pending	Word ladder - 2	Word Ladder II - LeetCode	https://youtu.be/0kPpYeLI4_Q	Word Ladder II Leetcode 126 Shortest Transformation Sequences with BFS	
Pending	Number of Islands	Number of Islands - LeetCode	https://youtu.be/_vh_eWEIEI8	Number of Islands Leetcode 200 Explained using BFS and DFS	
Pending	Number of Distinct Islands [dfs multisource]	Number of Distinct Islands Practice GeeksforGeeks	https://youtu.be/-mMmf3BAflc	Number of Distinct Islands Simple DFS solution in Python	
Pending	Bipartite Graph (DFS)	Is Graph Bipartite? - LeetCode	https://youtu.be/tXPMj5eEM	Is Graph Bipartite? Leetcode 785 Easy 2-Color DFS Solution in Python	
Pending	Cycle Detection in Directed Graph (DFS)	Directed Graph Cycle Practice GeeksforGeeks	https://youtu.be/BrxY15Gz2DY	Detect Cycle in a Directed Graph	
		15.3 Topo Sort based problems			
Pending	Topo Sort	Topological sort Practice GeeksforGeeks	https://youtu.be/hy7IVJNkgz	Topological Sort in Graph Simple DFS + Stack Solution in Python	
Pending	Kahn's Algorithm	Topological sort Practice GeeksforGeeks	https://youtu.be/5E89BR5Q-o	Topological Sort (Kahn's Algorithm) - Simple BFS Solution in Python	
Pending	Cycle Detection in Directed Graph (BFS)	Detect cycle in a directed graph Practice GeeksforGeeks	https://youtu.be/ZrwQoFY_4rE	Detect Cycle in a Directed Graph - Kahn's Algorithm (BFS Topological Check) in Python	
Pending	Course Schedule - I	Course Schedule - LeetCode	https://youtu.be/q2mqUGGPmwl	Course Schedule Leetcode 207 Kahn's Algorithm Walk-through in Python	
Pending	Course Schedule - II	Course Schedule II - LeetCode	https://youtu.be/q2mqUGGPmwl		
Pending	Find eventual safe states (BFS)	Find Eventual Safe States - LeetCode	https://youtu.be/bJLIEOPH4Nc	Find Eventual Safe States Leetcode 802 Kahn's Algorithm (BFS) Solution	
Pending	Find eventual safe states (DFS)	Find Eventual Safe States - LeetCode	https://youtu.be/tMIOLKzruRY	Find Eventual Safe States Leetcode 802 DFS Solution	
Pending	Alien dictionary	Alien Dictionary Practice GeeksforGeeks	https://youtu.be/yJRRRT13Bmo	[Do on your own]	
		15.4 Shortest Path related Problems			
Pending	Shortest Path in UC with unit weights	Shortest path in Undirected Graph Practice GeeksforGeeks	https://youtu.be/dR5EDT7do0	Shortest Path in an Undirected Graph with Unit Distance - Clear BFS Solution in Python	
Pending	Shortest Path in DAG	Shortest path in Directed Acyclic Graph Practice GeeksforGeeks	https://youtu.be/b8IB5UJ1p4U	Shortest Path in a Weighted Directed Acyclic Graph Topological-Sort + Relaxation Python	
Pending	Dijkstra's Algorithm [Using Priority Queue]	Implementing Dijkstra Algorithm Practice GeeksforGeeks	https://youtu.be/f0Xr7BfgZ2k	Dijkstra's Algorithm with a Priority Queue	
Pending	Dijkstra's Algorithm [Using Set]	Implementing Dijkstra Algorithm Practice GeeksforGeeks	https://youtu.be/2BLYNwYpMA	Dijkstra's Algorithm in Python Using a Set	
Pending	Why priority Queue is used in Dijkstra's Algorithm	Implementing Dijkstra Algorithm Practice GeeksforGeeks	https://youtu.be/kb4O-6cbWZg	[Watch Video]	
Pending	Print the Shortest Path	Shortest Path in Weighted undirected graph Practice GeeksforGeeks	https://youtu.be/loRgnK9uwCE	Printing the Actual Shortest Path with Dijkstra Algorithm	
Pending	Shortest path in a binary maze	Shortest Path in Binary Matrix - LeetCode	https://youtu.be/u6iPL2SQ0Fo	Shortest Path in a Binary Matrix Leetcode 1091 Dijkstra's Algorithm with Normal Queue (BFS)	
Pending	Path with minimum effort	Path With Minimum Effort - LeetCode	https://youtu.be/scW82V_eIFy	Path With Minimum Effort Leetcode 1631 Explained	
Pending	Cheapest flights within k stops	Cheapest Flights Within K Stops - LeetCode	https://youtu.be/8UJ4iq3LBWw	Cheapest Flights Within K Stops Leetcode 787 Easy BFS Approach in Python - Code and Debug - Blog	
Pending	Minimum steps to reach end from start by performing multiple jumps	Minimum Multiplications to reach End Practice GeeksforGeeks	https://youtu.be/H5p2kcyvQ6I	https://codeanddebug.in/blog/minimum-multiplications-to-reach-end-python-bfs/	
Pending	Number of ways to arrive at destination	Number of Ways to Arrive at Destination - LeetCode	https://youtu.be/7vFSTY4uGM	https://codeanddebug.in/blog/number-of-ways-to-arrive-at-destination-python/	
Pending	Bellman Ford Algorithm	Distance from the Source (Bellman-Ford Algorithm) Practice GeeksforGeeks	https://youtu.be/7C_wSRlEyoU	https://codeanddebug.in/blog/bellman-ford-algorithm/	
Pending	Floyd Warshal Algorithm	Floyd Warshall Practice GeeksforGeeks	https://youtu.be/LULHxlvOhGc	https://codeanddebug.in/blog/floyd-warshall-vs-dijkstra-all-pairs-shortest-paths/	
Pending	Find the city with the smallest number of neighbors in a threshold distance	Find the City With the Smallest Number of Neighbors at a Threshold Distance - LeetCode	https://youtu.be/KludkoXzOXY	https://codeanddebug.in/blog/find-the-city-with-the-smallest-number-of-neighbors-at-a-threshold-distance/	
		15.5 Minimum Spanning Tree and Disjoint Set			
Pending	Minimum Spanning Tree	Minimum Spanning Tree Practice GeeksforGeeks	https://youtu.be/Q-FY2CRhvk		
Pending	Prim's Algorithm	Minimum Spanning Tree Practice GeeksforGeeks	https://youtu.be/T2CNZcsYR7o		
Pending	Disjoint Set [Union by Rank]	Disjoint set (Union-Find) Practice GeeksforGeeks	https://youtu.be/E__JlsIBtik		
Pending	Disjoint Set [Union by Size]	Disjoint set (Union-Find) Practice GeeksforGeeks	https://youtu.be/E__JlsIBtik		

[illegible]